# PRISM surveillance report

PatientID_0003_Timepoint_1 | scan day 90 from diagnosis | early post-RT (0-84 days after the end of radiotherapy)

### Risk of documented progression within 120 days

## 0.50 (95% interval 0.26-0.69)

Tier: middle third of development-cohort predictions (cut points 0.30 / 0.53). 80% conformal set: both outcomes (progression and no progression).

### Referral flags

**[!] early post-RT window: scan 25 days after the end of RT, inside the 84-day early post-RT window where treatment effects can mimic progression**

**[!] out of distribution: MGMT status unknown**

**[!] uncertain: the 80% conformal prediction set contains both outcomes**

Action: flagged for clinician review

### Contributions (exact additive log-odds; positive = towards higher risk)

Intercept -0.20 clinical +0.36 timing -0.41 treatment +0.26 = logit +0.01. Imaging: not part of this model.

-0.17 scan number = 1 (timing)
-0.14 within 84 days after RT end: yes (timing)
+0.12 male sex: no (clinical)
+0.11 day of first surgery = -5 (treatment)
+0.11 previous brain tumour: no (treatment)

### Inputs as of the scan day (source: MU clinical workbook)

Age 57, female; IDH wildtype; MGMT unknown; RT started yes, completed yes; days since RT end 25; chemotherapy started yes. All 23 inputs known by the scan day: yes (each with source column and as-of day in the record).

### Volumetric change (indicative, not RANO)

Enhancing 33.8 cc; FLAIR hyperintensity 44.3 cc.

First scan: no prior or nadir.

### Summary (local LLM; every sentence cites record fields; checked deterministically)

This scan was performed 90 days from diagnosis, 25 days after the end of radiotherapy, during the early post-RT phase of care.

The model estimates a 50% risk of documented progression within 120 days, with a 95% interval ranging from 26% to 69%, placing the patient in the middle third of risk.

The 80% conformal prediction set includes both progression and no progression outcomes, indicating uncertainty in the specific trajectory.

The scan is flagged for clinician review because it falls within the early post-RT window where treatment effects can mimic progression, MGMT status is unknown, and the prediction set is uncertain.

Indicative imaging volumes, which are not a RANO assessment, show an enhancing volume of 33.8 cc and FLAIR hyperintensity of 44.3 cc.

This tool is intended for decision support alongside clinical and radiological assessment and does not provide a diagnosis.

Faithfulness check: 6/6 claims supported; active flags mentioned: all; forbidden content: none.

### Limitations

Model discrimination in development cross-validation AUC 0.70 (single institution, MU-Glioma-Post); the risk is the probability of documented progression within 120 days, not a diagnosis of progression or pseudoprogression; volumes are indicative, not a RANO assessment.